Prayas JEE 3.0 2025

Mathematics

Trigonometric Equation

DPP: 1

Q1 Find the principal solution of the following equations $\sec x=2$

(A) $\frac{\pi}{6}, \frac{\pi}{4}$
(B) $\frac{\pi}{3}, \frac{5\pi}{3}$
(C) $\frac{7\pi}{3}, \frac{9\pi}{4}$
(D) None of these

Q2 If $\cot\theta-\tan\theta=2$, then find principal solution of θ.

(A) $\frac{\pi}{4}, \frac{5\pi}{4}$
(B) $\frac{\pi}{8}, \frac{5\pi}{8}$
(C) $-\frac{\pi}{4}, -\frac{5\pi}{4}$
(D) $-\frac{\pi}{8}, \frac{5\pi}{8}$

Q3 Find the general solution of the following equation: $2\sin\theta+1=0$

(A) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{\pi}{6}$
(B) $\mathrm{n}\pi+(-1)^{\mathrm{n}+1}\frac{\pi}{6}$
(C) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{5\pi}{6}$
(D) $\mathrm{n}\pi+(-1)^{\mathrm{n}}\frac{\pi}{3}$

Q4 Write the general solution of the trigonometric equation $4\tan^2\theta=3\sec^2\theta$

(A) $n\pi\pm\frac{\pi}{6}$
(B) $\mathrm{n}\pi\pm\frac{\pi}{3}$
(C) $\mathrm{n}\pi\pm\frac{5\pi}{6}$
(D) $\mathrm{n}\pi\pm\frac{2\pi}{3}$

Q5 The solution of $(\sec\theta+1)=(2+\sqrt{3})\tan\theta$ $(0<\theta<2\pi)$ are

(A) $\frac{\pi}{6}, \pi$
(B) $\frac{\pi}{3}, \frac{\pi}{4}$
(C) $\pi/6, 2\pi/3$
(D) None of these

Q6 Find Principle solution of $\sin 2x+2\sin x+2\cos x+1=0$

(A) $\frac{\pi}{4}$
(B) $\frac{3\pi}{4}$
(C) $\frac{2\pi}{3}$
(D) $\frac{\pi}{2}$

Q7 If $3\left(\sec^2\theta+\tan^2\theta\right)=5$, then principle value of θ.

(A) $\frac{\pi}{3}\&\frac{2\pi}{3}$
(B) $\frac{\pi}{6}\&\frac{5\pi}{6}$
(C) $0\&\pi$
(D) $\frac{\pi}{2}\&\frac{3\pi}{2}$

Q8 If $\sin^2\theta-2\cos\theta+\frac{1}{4}=0$, then the general value of θ is

(A) $\mathrm{n}\pi\pm\frac{\pi}{3}$
(B) $2n\pi\pm\frac{\pi}{3}$
(C) $2\mathrm{n}\pi\pm\frac{\pi}{6}$
(D) $\mathrm{n}\pi\pm\frac{\pi}{6}$

Q9 If $\cos\theta=-\frac{1}{2}$ and $\tan\theta=\sqrt{3}$, then find the most general value of θ satisfying both the equations.

(A) $2n\pi+\frac{\pi}{3}$
(B) $2n\pi+\frac{2\pi}{3}$
(C) $2\mathrm{n}\pi+\frac{4\pi}{3}$
(D) None

Q10 If $\tan\theta+\tan 2\theta+\tan 3\theta=\tan\theta\tan 2\theta\tan,$ 3θ then the general value of $\boldsymbol{\theta}$ is

(A) $n\pi$
(B) $\frac{\mathrm{n}\pi}{6}$ (where *n* is even)
(C) $\mathrm{n}\pi\pm\frac{\pi}{3}$
(D) $\frac{\mathrm{n}\pi}{2}$

Answer Key

Q1	(B)
Q2	(B)
Q3	(B)
Q4	(B)
Q5	(A)
Q6	(B)
Q7	(B)
Q8	(B)
Q9	(C)
Q10	(B)

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